

Lecture Objectives and Pre-Requisites

Key Takeaways

The key takeaways from this lecture are:

- Limitations of static models
- How ARCH/GARCH address the above problem
- Impact of transaction costs on strategy performance
- Using automated parameter search
- Integrated ARIMA+GARCH models for more robust forecasting frameworks.

Prerequisites

You need to have attended the ASQ-01 lecture or at least gone through its recording and class files.

To follow along and implement the concepts covered in the ASQ-02 lecture, ensure that you have the following installed in your environment:

- **numpy, pandas:** numerical/data operations
- **matplotlib, seaborn:** data visualization
- **yfinance:** financial data
- **statsmodels:** ARIMA, ARCH, diagnostics
- **arch:** advanced ARCH/GARCH modeling
- **sktime:** automated model selection
- **datetime, time:** handling dates and timing execution

Please refer to the documentation for each library if you encounter any issues during installation. In case you have trouble installing one or more of them, it'll be a minor annoyance for you during the lecture (since you won't be able to run some of the code). However, you should still be able to follow most of the material covered in the lecture.



Recommended Time for the Session & Related Coursework - 9 hours

Additional Read (Blog):

<https://blog.quantinsti.com/garch-gjr-garch-volatility-forecasting-python/>